From Raw Taxi Trips to Route Intelligence: A Two-Column Research Report on a Predictive Data-Wrangling Pipeline Using NYC TLC Yellow Taxi Data

Sunain Mushtaq
SIT220 Data Wrangling
Deakin University
Student ID: S224089221
s224089221@deakin.edu.au

***Abstract*—This paper presents an end-to-end route-intelligence pipeline using New York City Taxi and Limousine Commission (NYC TLC) Yellow Taxi trip records from January to March 2026. The work is framed as a data-wrangling and predictive-modelling study rather than as a model-only exercise. Raw monthly Parquet files are loaded, validated, cleaned, joined with taxi-zone metadata, transformed into temporal and route-level features, aggregated into route-hour corridors, scored using an interpretable route relevance index, and then used for predictive modelling. The raw dataset contained 11,077,206 trips and the final cleaned dataset contained 10,487,427 trips after removing invalid timestamps, implausible durations, impossible distances, negative payment values, out-of-scope pickup dates, duplicate records, and extreme speeds. Route demand was strongly concentrated around repeated Manhattan movements and airport-related corridors. A Ridge regression baseline predicted trip duration with a mean absolute error (MAE) of 5.97 minutes and R^2 of 0.631, while a HistGradientBoosting model reduced MAE to 3.79 minutes and increased R^2 to 0.821. A secondary Ridge model predicted `fare_amount` with MAE of $3.46 and R^2 of 0.845. The study shows how raw operational taxi data can be converted into a reproducible analytical layer for route relevance, travel-time prediction, fare behaviour interpretation, and cautious business feasibility analysis.**

***Index Terms*—data wrangling, NYC TLC, Yellow Taxi, route intelligence, feature engineering, Ridge regression, HistGradientBoosting, urban mobility analytics**

I. Introduction

NYC Yellow Taxi trip records provide a large-scale operational record of completed taxi movements across New York City. Each trip includes pickup and drop-off timestamps, taxi-zone identifiers, trip distance, fare components, payment indicators, passenger count, and related operational attributes. These fields make the data valuable for urban mobility analysis, but the raw records do not directly state which taxi corridors are consistently important, which route patterns are reliable, or which trip features support useful prediction.

This report presents the notebook as a continuous research pipeline. The analysis begins with official TLC monthly Parquet files and gradually transforms them into an interpretable route-intelligence layer. The core analytical unit is a route-hour combination: pickup zone, drop-off zone, pickup day of week, and pickup hour. This design is important because taxi demand is both spatial and temporal. A zone-to-zone route may be useful during weekday evening periods but weak during early-morning windows. Treating all trips as isolated records would hide this recurring operational structure.

The central research question is: *How can NYC Yellow Taxi trip records be transformed into a route-level predictive pipeline that estimates trip duration, fare behaviour, and route relevance while providing a transparent foundation for future customer-driver optimisation and business feasibility analysis?*

Three sub-questions guide the analysis. First, which pickup-to-drop-off zone pairs recur reliably across specific hours and days? Second, which engineered features best explain duration and fare behaviour? Third, how can route relevance, predicted duration, and predicted fare be used as proxy indicators for passenger reliability and driver-facing value without claiming that the TLC data directly measures satisfaction or platform profitability?

II. Related Framing and Contribution

The contribution is practical and pipeline-oriented. The work does not attempt to build a live dispatch system, and it does not claim to observe customer ratings, driver ratings, driver payout, live driver availability, rejected requests, or platform costs. Instead, it uses the observed structure of completed Yellow Taxi trips to construct a defensible analytical foundation. Route reliability is treated as a possible passenger-facing proxy, repeated demand and fare potential are treated as possible driver-facing value proxies, and profit-style calculations are kept as scenario-based extensions.

The project is therefore positioned as a data-wrangling research report with predictive modelling as one stage of the workflow. Cleaning, validation, feature design, route aggregation, and interpretation matter as much as the final model scores. This is aligned with the SIT220 objective of demonstrating how raw operational data can be transformed into reusable analytical outputs.

III. Data Scope

The study uses official NYC TLC Yellow Taxi trip records for January, February, and March 2026, with the TLC taxi-zone lookup table used for metadata integration. The three-month window creates a dataset large enough to show realistic data quality, aggregation, and modelling challenges, while still remaining feasible for a notebook-based workflow.

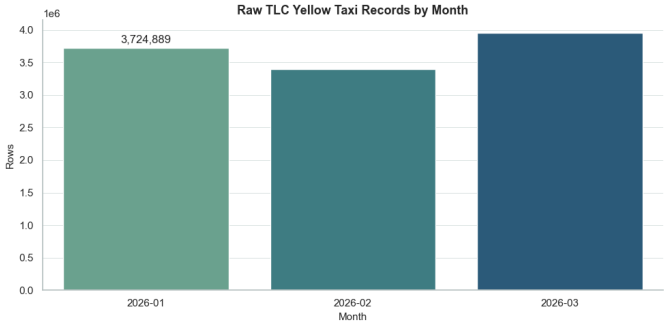


Fig. 1: Raw Yellow Taxi record counts by month for January to March 2026.

The raw input contained 11,077,206 trip records. The selected files were treated consistently through the same validation and cleaning functions so that downstream route-hour aggregation and modelling would not be affected by month-specific processing differences.

IV. Pipeline Design

The workflow was structured as a sequence of dependent stages. Each stage produces an output that becomes the input for the next stage. This prevents the report from becoming a collection of disconnected code cells and makes the modelling results traceable back to the raw data preparation decisions.

V. Validation and Cleaning

The raw TLC dataset is valuable because it contains detailed trip, location, distance, and payment variables. However, the validation stage shows why strict preprocessing is required before route scoring or modelling. In a dataset of this size, even a small invalid share can represent hundreds of thousands of records and can distort route counts, travel-time distributions, fare estimates, and model learning.

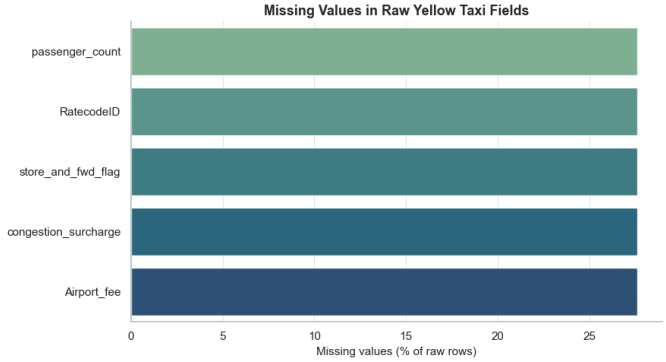


Fig. 2: Missing values in key raw Yellow Taxi fields before cleaning.

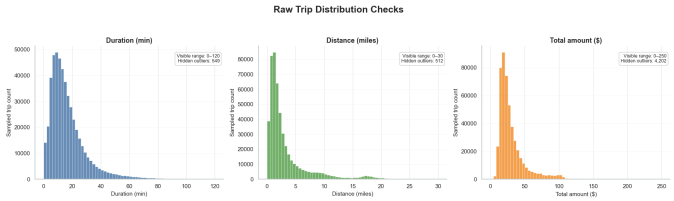


Fig. 3: Raw trip distribution checks used to identify skew, implausible values, and outlier-sensitive variables.

The cleaning logic removed records with invalid pickup/drop-off ordering, implausible durations, impossible or near-zero trip distances, negative payment values, pickup timestamps outside the selected January–March 2026 period, exact duplicates, and extreme average speeds. Passenger-count records coded as zero were not globally removed because they still carry useful timestamp, location, distance, and payment information for route-hour aggregation. Passenger count is treated later as a weak modelling control rather than as a core route relevance variable.

TABLE II: Cleaning summary for the selected TLC records.

Dataset stage	Rows	Share
Raw records	11,077,206	100.00%
Cleaned records	10,487,427	94.68%
Removed records	589,779	5.32%

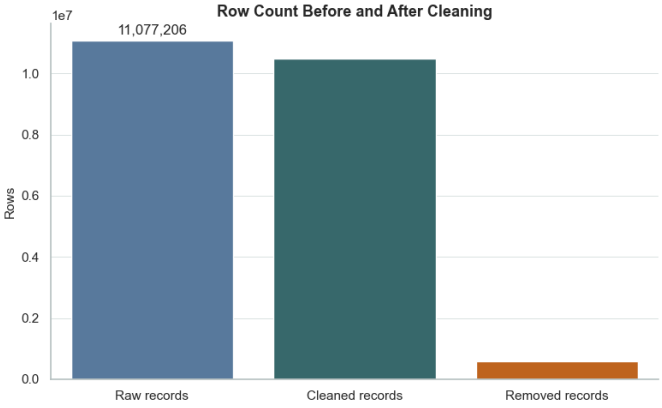


Fig. 4: Row count before and after the cleaning pipeline.

VI. Metadata Integration and Feature Engineering

The cleaned records were joined with the TLC taxi-zone lookup table to convert numeric pickup and drop-off IDs into readable zone and borough labels. This made the final outputs interpretable. For example, a route ID can be translated into a corridor such as Upper East Side South to Upper East Side North rather than remaining as a pair of numeric location identifiers. After joining, the route table contained 10,487,427 rows. Missing pickup-zone names occurred in 11,684 rows and missing drop-off-zone names occurred in 14,489 rows, which is small relative to the full cleaned dataset. The numeric location IDs remain available even where zone names are missing.

TABLE I: End-to-end analytical pipeline used in the notebook.

Stage	Role in the research workflow
Raw data acquisition	Loads the January–March 2026 Yellow Taxi Parquet files and defines the study scope.
Schema inspection	Checks available columns, data types, row counts, missingness, and variables required for modelling.
Validation	Detects invalid timestamps, impossible distances, negative fares, duplicate rows, out-of-scope dates, and speed outliers.
Cleaning	Creates a reproducible cleaned trip-level dataset suitable for aggregation and modelling.
Zone metadata join	Converts pickup and drop-off location IDs into readable zone and borough labels.
Feature engineering	Creates temporal, route, distance, fare, speed, movement, and reliability features.
Route-hour aggregation	Groups trips by pickup zone, drop-off zone, pickup day, and pickup hour.
Route relevance scoring	Assigns a continuous operational score based on demand, fare value, and reliability.
Predictive modelling	Compares Ridge and HistGradientBoosting models for duration; adds a secondary Ridge fare model.
Business proxy analysis	Interprets route relevance and model outputs as cautious proxies for reliability, driver value, and scenario margins.

Feature engineering was organised around the logic of the pipeline. Temporal features captured pickup hour, day of week, weekend status, and month. Route features captured pickup zone, drop-off zone, route ID, and borough movement. Distance and speed features supported duration modelling and quality checks. Fare features supported fare interpretation and the later business proxy layer. Route-level features captured recurring demand, duration reliability, fare behaviour, and consistency.

VII. Exploratory Route Analysis

Exploratory analysis confirms that taxi demand is highly structured by time. Pickup demand rises after the early-morning low, grows through the day, and remains strong during afternoon and evening periods. This supports the decision to use route-hour combinations rather than static pickup-to-drop-off pairs.

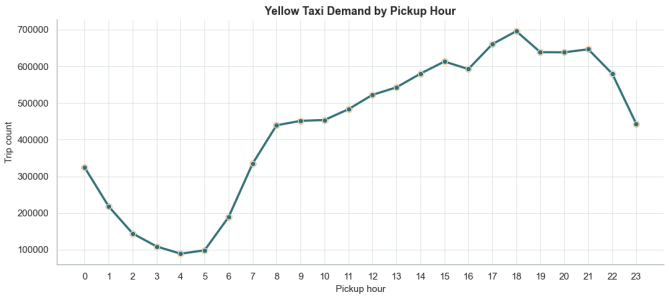


Fig. 5: Yellow Taxi demand by pickup hour after cleaning.

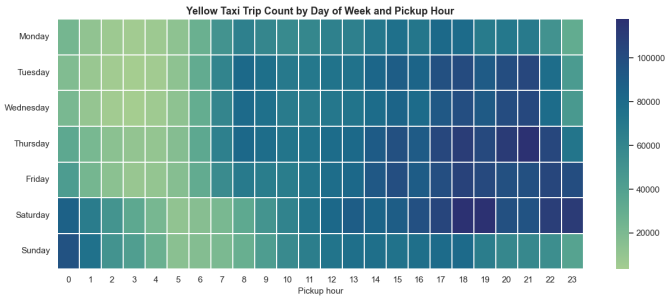


Fig. 6: Trip-count heatmap by day of week and pickup hour. The pattern shows that taxi demand varies strongly across both temporal dimensions.

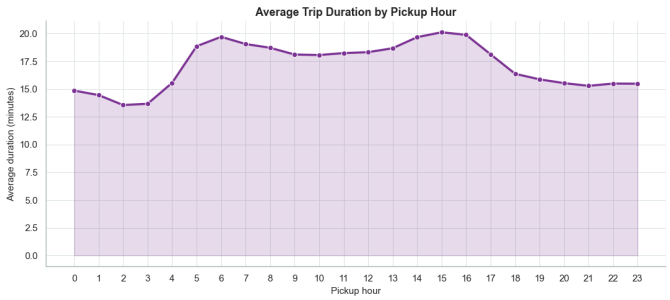


Fig. 7: Average trip duration by pickup hour. Duration varies across the day, reinforcing the need for temporal features in prediction.

The top recurring route pairs are dominated by dense Manhattan movements. These short, repeated zone-to-zone corridors are operationally different from long airport movements. A route-intelligence layer must therefore account for both frequency and route quality, not just raw volume.

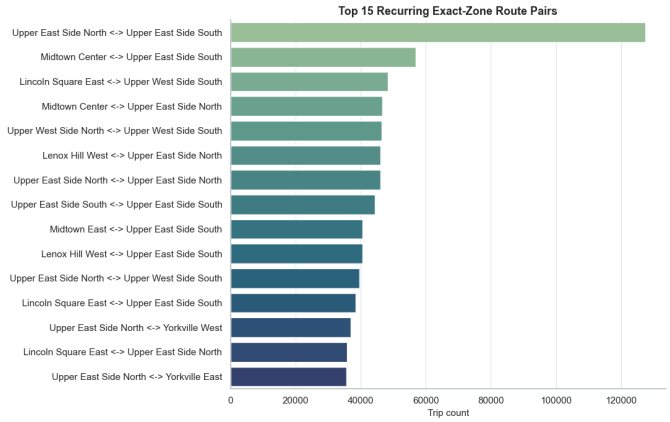


Fig. 8: Top recurring exact pickup-to-drop-off taxi-zone pairs.

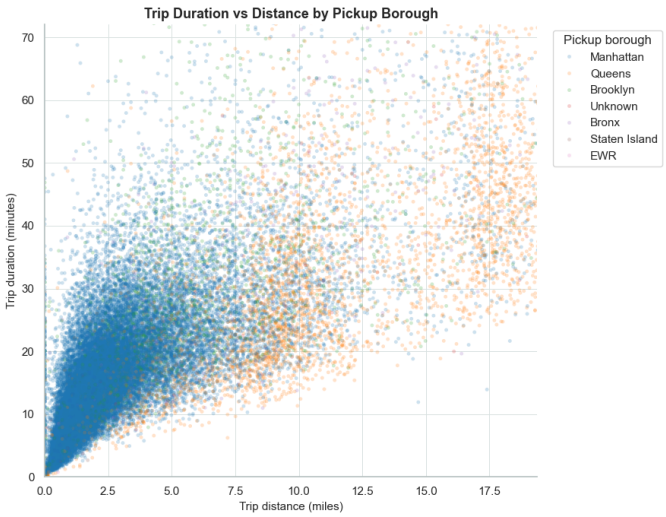


Fig. 9: Trip duration versus distance by pickup borough. The relationship is strong but non-linear, motivating a non-linear duration model.

VIII. Route-Hour Aggregation

The notebook groups trips by pickup day of week, pickup hour, pickup zone, and drop-off zone. This produces the route-hour table that underpins the rest of the research. The route key is not a fixed public transport line; it is a recurring taxi corridor inferred from repeated completed trips.

Route-hour aggregation reduces millions of individual trips into interpretable operational units. It also allows each corridor to be described using frequency, average duration, duration variability, average fare, and recurrence. The distribution of route-hour trip counts is highly skewed, with many rare route-hour combinations and a smaller number of heavily repeated corridors.

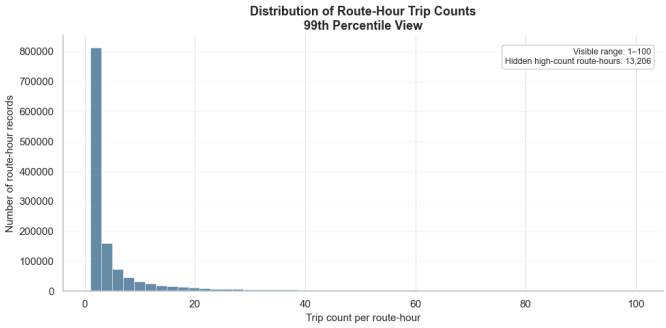


Fig. 10: Distribution of route-hour trip counts, showing that route activity is concentrated rather than evenly distributed.

IX. Route Relevance Scoring

Raw trip count alone is not enough to define a useful taxi route. A high-volume route may have weak fare value or unstable duration. A high-fare route may occur too rarely to be operationally useful. The route relevance score addresses this by combining multiple interpretable components into a continuous index.

The notebook uses a balanced manually defined score based on frequency, fare value, and duration reliability. The initial weighting prioritises repeated demand while still rewarding commercial value and stable travel time:

$$R = 0.45F + 0.25V + 0.30L, \tag{1}$$

where R is route relevance, F is normalised frequency, V is normalised fare value, and L is duration reliability. The score is not treated as a hidden target learned by a model. It is an interpretable operational index that can later be adjusted for passenger-focused, driver-focused, or platform-focused priorities.

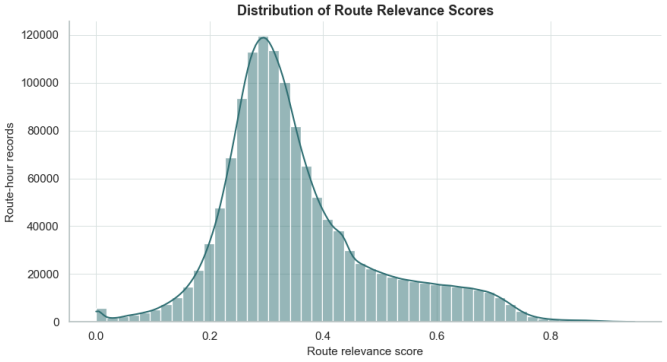


Fig. 11: Distribution of route relevance scores across route-hour combinations.

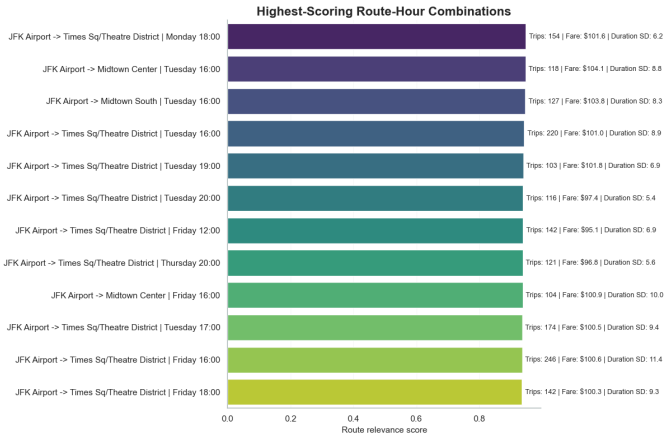


Fig. 12: Highest-scoring route-hour combinations under the balanced relevance score. Airport-related and recurring high-value corridors appear strongly.

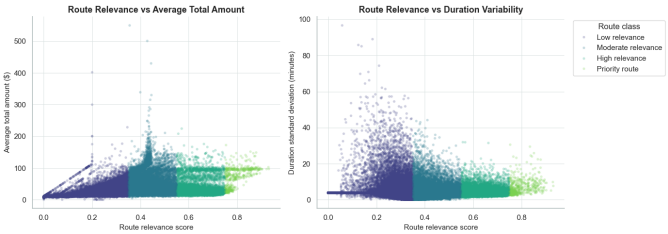


Fig. 13: Route relevance compared with fare value and duration reliability. The score captures more than simple frequency.

Spatially, high relevance is not randomly distributed. Stronger values appear around dense Manhattan pickup zones and major transport hubs, which reflects the overlap between repeated demand, fare value, and route stability.

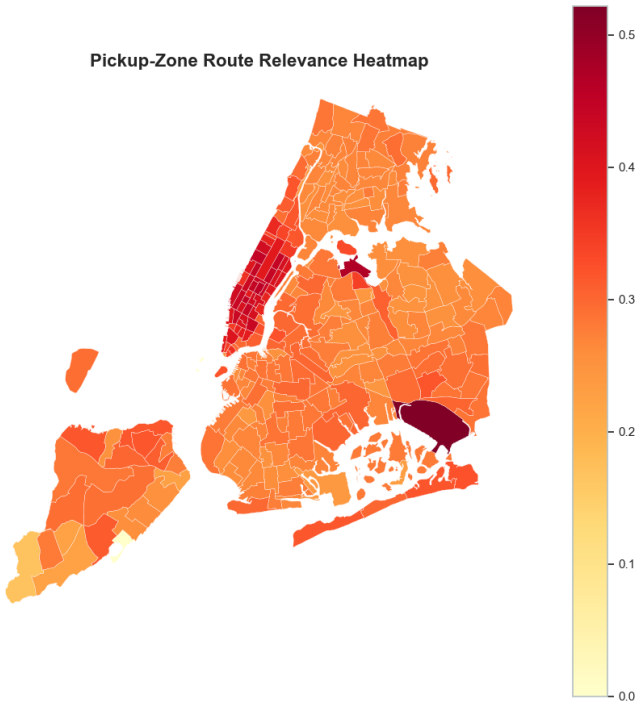


Fig. 14: Pickup-zone route relevance heatmap. Darker zones represent stronger average route relevance.

X. Modelling Problem Definition

The modelling stage appears after wrangling, feature engineering, and route scoring. This order matters because the prediction tasks should be based on features that are clearly defined, validated, and interpretable. The primary target is trip duration in minutes. Duration is a defensible target because it is directly derived from timestamps and strongly linked to passenger-facing reliability and driver planning. The secondary targets are `fare_amount` and `total_amount`. These support business interpretation, but they are treated carefully because some fare components may be rule-based or determined after trip completion.

The duration models use a consistent train/test split and the same prepared feature matrix. Ridge regression is used as the baseline because it provides a simple regularised linear benchmark. HistGradientBoosting is used as the stronger model because it can capture non-linear relationships between distance, pickup hour, route structure, borough movement, and route-level behaviour.

XI. Duration Prediction Results

The HistGradientBoosting model generalised well during training. The training and validation curves converged without showing strong overfitting, supporting the choice of the stronger model for the final comparison.

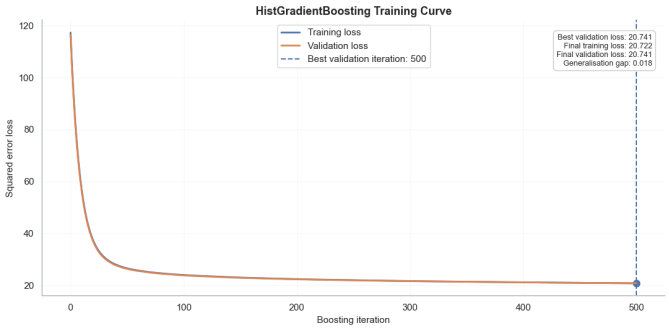


Fig. 15: HistGradientBoosting training curve for trip-duration prediction.

TABLE III: Duration model comparison on the held-out test set.

Model	MAE	RMSE	R^2
Ridge baseline	5.972	9.333	0.631
HistGradientBoosting	3.787	6.492	0.821

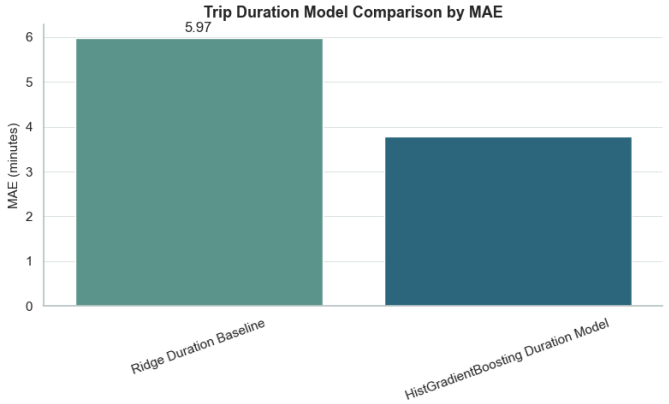


Fig. 16: Trip-duration model comparison by MAE. The non-linear model reduces average duration error substantially.

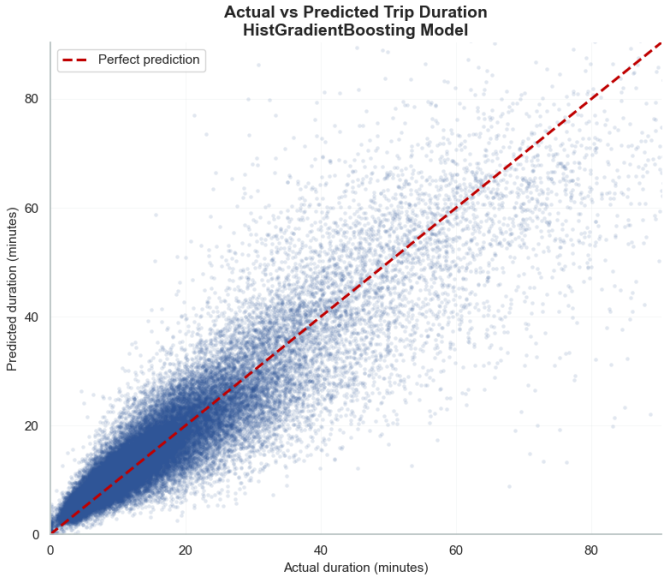


Fig. 17: Actual versus predicted trip duration for the HistGradientBoosting model.

The Ridge baseline achieved MAE of 5.97 minutes, RMSE of 9.33 minutes, and R^2 of 0.631. The HistGradientBoosting model reduced MAE to 3.79 minutes, reduced RMSE to 6.49 minutes, and increased R^2 to 0.821. This is a clear improvement and confirms that engineered time, distance, route, and borough features contain predictive signal. The result also supports the earlier exploratory finding that taxi duration is not purely linear.

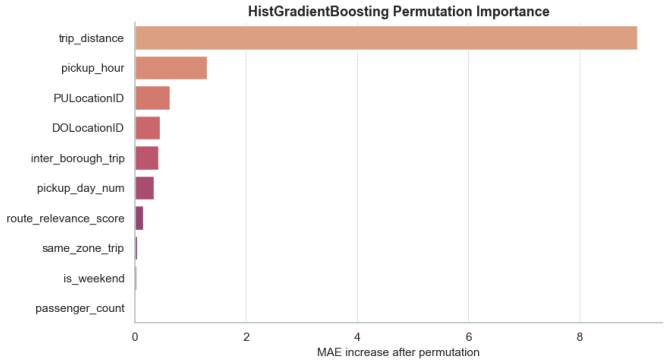


Fig. 18: Permutation importance for the HistGradientBoosting duration model. Trip distance dominates, while pickup hour, route IDs, borough movement, and relevance features add supporting signal.

Permutation importance shows that trip distance is the strongest predictor, which is expected for duration modelling. Pickup hour, location features, borough movement, weekday status, and route relevance also contribute. Passenger count contributes very little, which supports treating it as a weak control rather than a major explanatory feature.

XII. Fare Prediction Results

Fare prediction was included as a secondary modelling task. A two-target Ridge model estimated `fare_amount` and `total_amount`. The `fare_amount` target is closer to the metered fare component, while `total_amount` includes additional charges and tip behaviour that may be harder to estimate from leakage-conscious features.

TABLE IV: Secondary fare model results.

Target	MAE	RMSE	R^2
`fare_amount`	\$3.464	\$7.106	0.845
`total_amount`	\$5.212	\$9.562	0.819

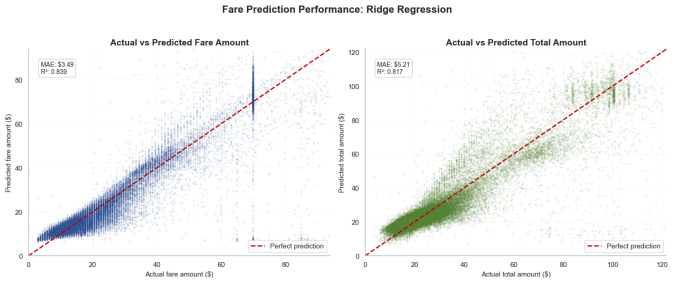


Fig. 19: Actual versus predicted fare amount and total amount using Ridge regression. Fare amount is easier to estimate than total amount.

The fare model supports passenger-facing price interpretation and later business scenario analysis, but it is not the primary contribution of the project. Fare prediction must be interpreted carefully because some charges are deterministic, some are rule-based, and some are partly determined after the trip. Duration prediction remains the stronger main target because it is directly linked to reliability and can be predicted more defensibly from pre-trip and route-level features.

XIII. Business Proxy Layer

The business layer is treated as a research extension, not as a direct measurement of profitability or satisfaction. The TLC data contains passenger payments, but it does not contain driver payout, platform operating costs, rejected requests, ratings, or live wait times. Therefore, the notebook uses transparent assumptions and proxy logic.

Passenger-facing reliability is represented through predicted duration and duration stability. Driver-facing value is represented through repeated demand, fare potential, and route relevance. Platform-style margin is explored through scenario assumptions such as passenger payment minus assumed driver share and operating cost.

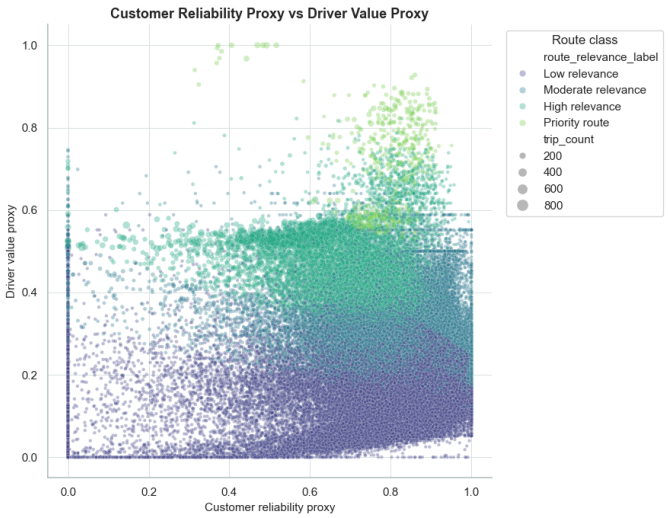


Fig. 20: Customer reliability proxy versus driver value proxy. Route relevance supports the interpretation of passenger-driver trade-offs.

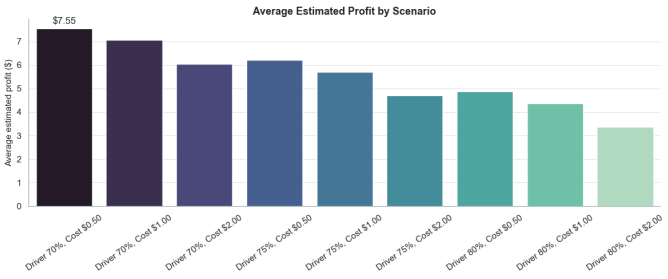


Fig. 21: Average estimated profit under scenario assumptions. The values should be interpreted as exploratory PnL proxies, not observed platform profit.

The proxy layer is useful because it connects the technical pipeline to a broader decision problem. A route can be frequent but unreliable, high fare but rare, or reliable but commercially weak. The route relevance score gives a structured way to compare these trade-offs. Future work could expose separate customer-weighted, driver-weighted, and platform-weighted relevance scores instead of one fixed balanced score.

XIV. Key Insights

The first major insight is that route demand is concentrated rather than evenly distributed. The dataset contains tens of thousands of exact pickup-to-drop-off route IDs, but high operational relevance is concentrated in a smaller set of repeated Manhattan corridors and airport-related movements. This justifies the route-hour design because the same route can have different operational value depending on day and hour.

The second insight is that strict validation is not optional. Removing 5.32% of raw records produced a cleaner dataset while preserving the broader structure of taxi activity. Invalid durations, impossible distances, negative fares, extreme speeds, and out-of-scope timestamps would have distorted route relevance, model training, and business interpretation if left untreated.

The third insight is that route relevance adds more value than raw trip count. A useful route is not only common; it should also be reliable, repeated, and commercially meaningful. The score makes this explicit and interpretable.

The fourth insight is that the non-linear duration model is meaningfully stronger than the linear baseline. HistGradientBoosting improved all three evaluation metrics and produced a more accurate duration estimator because urban travel time depends on interactions between distance, time, location, borough movement, and route behaviour.

The fifth insight is that fare modelling is useful but secondary. The model predicts metered fare amount well, but total amount is harder because it includes additional components such as tips and surcharges. This reinforces the decision to keep trip duration as the main prediction target.

XV. Limitations

Several limitations remain. First, the study uses completed Yellow Taxi trips only. It does not observe unserved demand, rejected requests, live driver supply, customer wait time, or platform dispatch decisions. Second, the analysis covers only January to March 2026, so seasonal and long-run changes are outside the current scope. Third, route relevance weights are manually chosen and should be treated as transparent assumptions rather than learned ground truth. Fourth, fare-related modelling can be affected by deterministic fare rules and post-trip components. Fifth, the route-hour table can become sparse when exact pickup and drop-off zones are combined with hour and day filters.

These limitations do not invalidate the pipeline, but they define the boundary of what the report can claim. The work demonstrates route-intelligence construction and predictive modelling from historical trips. It does not claim to be a live routing optimiser or a complete platform-profit model.

XVI. Future Work

Future work can extend the pipeline in five directions. The first direction is longer temporal coverage, such as adding additional months to capture seasonal changes and improve route recurrence estimates. The second is spatial generalisation, where exact taxi zones could be grouped into boroughs, airport corridors, Midtown corridors, or manually defined service regions to reduce sparsity. The third is model tuning, especially for HistGradientBoosting hyperparameters and possible tree-based alternatives. The fourth is fare decomposition, separating deterministic fare components from uncertain components such as tolls, tips, and time-varying charges. The fifth is dashboard reporting, where route relevance, time filters, borough filters, fare value, and reliability could be explored interactively using Plotly, Bokeh, or Folium.

XVII. Conclusion

This project developed an end-to-end route-intelligence pipeline using NYC TLC Yellow Taxi records from January to March 2026. The notebook began with more than 11 million raw trip records and produced a cleaned analysis-ready

dataset of 10,487,427 trips after removing invalid timestamps, implausible durations, impossible distances, negative fares, out-of-scope pickup dates, duplicate records, and extreme speeds. This cleaning stage is a core contribution because the project demonstrates how raw transport data can be validated, transformed, and prepared for reliable analysis.

The final pipeline connects raw Parquet loading, schema inspection, validation, cleaning, taxi-zone metadata integration, feature engineering, route-hour aggregation, route relevance scoring, predictive modelling, evaluation, and business proxy analysis. This structure converts raw operational records into an interpretable analytical layer rather than simply feeding unprocessed records into a model.

A key outcome is the route-hour intelligence layer. By grouping trips by pickup zone, drop-off zone, pickup day, and pickup hour, the analysis identifies recurring taxi corridors and shows how their usefulness changes across time. High-priority route-hour combinations are concentrated around repeated Manhattan movements and airport-related corridors. The route relevance score makes this interpretation practical by combining frequency, fare potential, and duration reliability into a balanced operational index.

The predictive modelling stage strengthens the report by showing that engineered features can estimate ride outcomes. Ridge regression provides a transparent baseline, while HistGradientBoosting captures non-linear duration behaviour and clearly improves performance. The secondary fare model supports price interpretation but remains a supporting task. Overall, the report demonstrates a coherent data-wrangling and predictive-modelling workflow that can support route analysis, operational interpretation, and future business feasibility modelling while staying grounded in what the TLC dataset actually contains.

REFERENCES

[1] New York City Taxi and Limousine Commission, "TLC Trip Record Data," NYC TLC, 2026.

[2] New York City Taxi and Limousine Commission, "Taxi Zone Lookup Table," NYC TLC, 2026.

[3] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[4] W. McKinney, "Data structures for statistical computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 56–61.